Validation-Oriented CFD Investigation of Conjugate Heat Transfer in NASA C3X Turbine Guide Vane *(Ongoing Study)*

This self-initiated research project focuses on the numerical investigation of conjugate heat transfer in a high-pressure turbine guide vane based on the NASA C3X experimental configuration. The main objective of the study is to understand the aerothermal behaviour of turbine components operating under high-temperature gas flow conditions and to evaluate the effectiveness of internal cooling strategies used to maintain safe blade temperatures in gas turbine engines.

A two-dimensional periodic cascade computational domain was developed by reconstructing the NASA C3X turbine vane geometry from published experimental data, followed by careful definition of inlet, outlet, and periodic boundary conditions to represent realistic cascade flow behaviour. High-quality meshing strategies combining structured and unstructured elements were applied with appropriate boundary layer refinement to accurately capture near-wall heat transfer and flow gradients.

The compressible flow field was simulated using suitable turbulence modelling approaches, and conjugate heat transfer analysis was performed to account for the interaction between internal cooling flow, solid heat conduction within the vane material, and external hot gas flow. Numerical convergence behaviour and mesh sensitivity considerations were carefully monitored to ensure solution stability and reliability.

The study involves detailed analysis of key aerodynamic and thermal flow parameters, including Mach number distribution, static and total pressure variation, and temperature gradients along the vane surface. Validation of computational predictions is being carried out through comparison with published NASA experimental data for selected operating conditions. This validation-oriented approach helps improve confidence in numerical results and supports deeper understanding of turbine cooling performance.

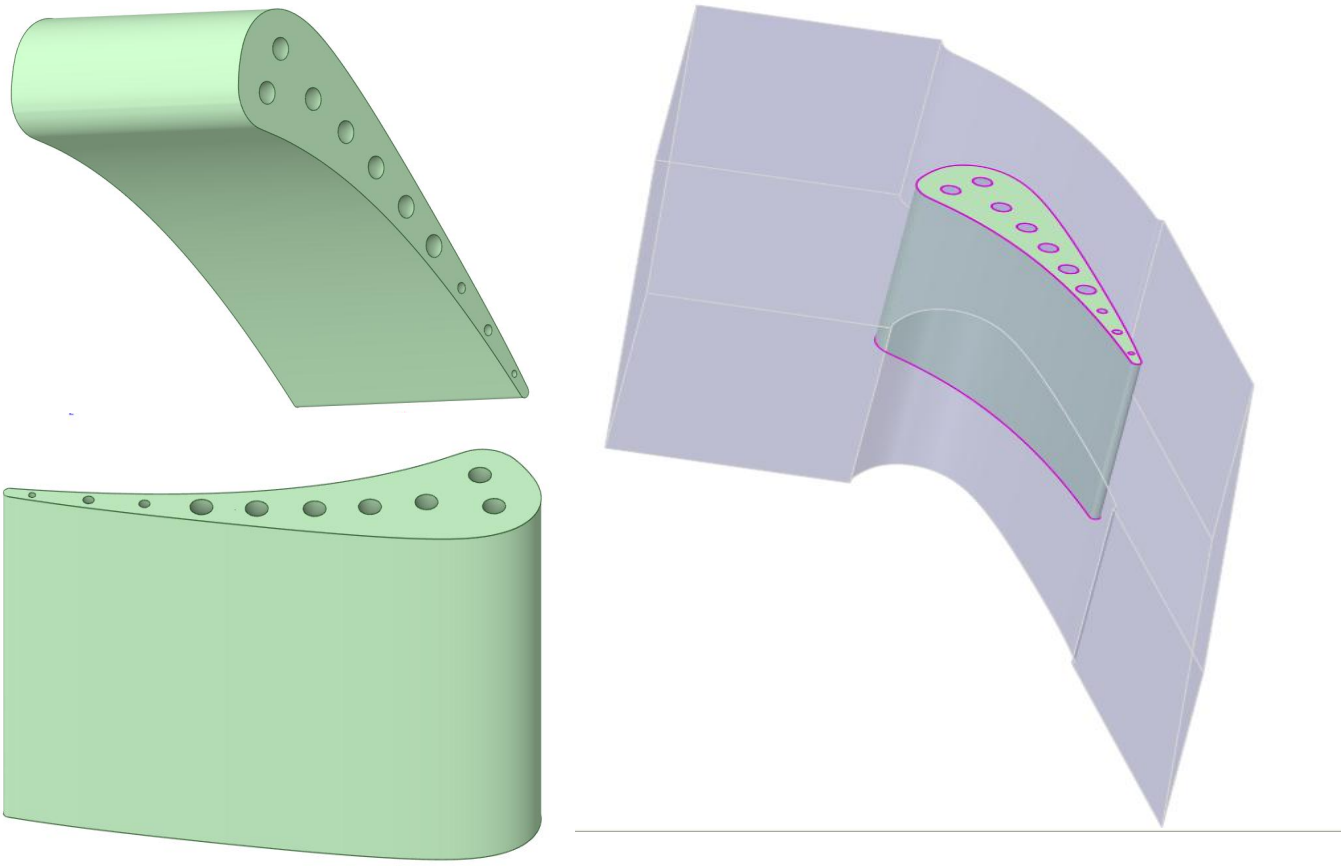
Through this ongoing work, I have developed strong skills in computational fluid dynamics, turbomachinery aerothermal analysis, geometry modelling, mesh generation, numerical validation techniques, and performance interpretation of high-

temperature propulsion components. Further work is aimed at improving prediction accuracy and investigating the influence of cooling flow parameters on turbine thermal management effectiveness and aerodynamic performance.

Computational results from the ongoing study are presented below to illustrate modelling approach and key aerodynamic and thermal flow features.

FIGURE 1

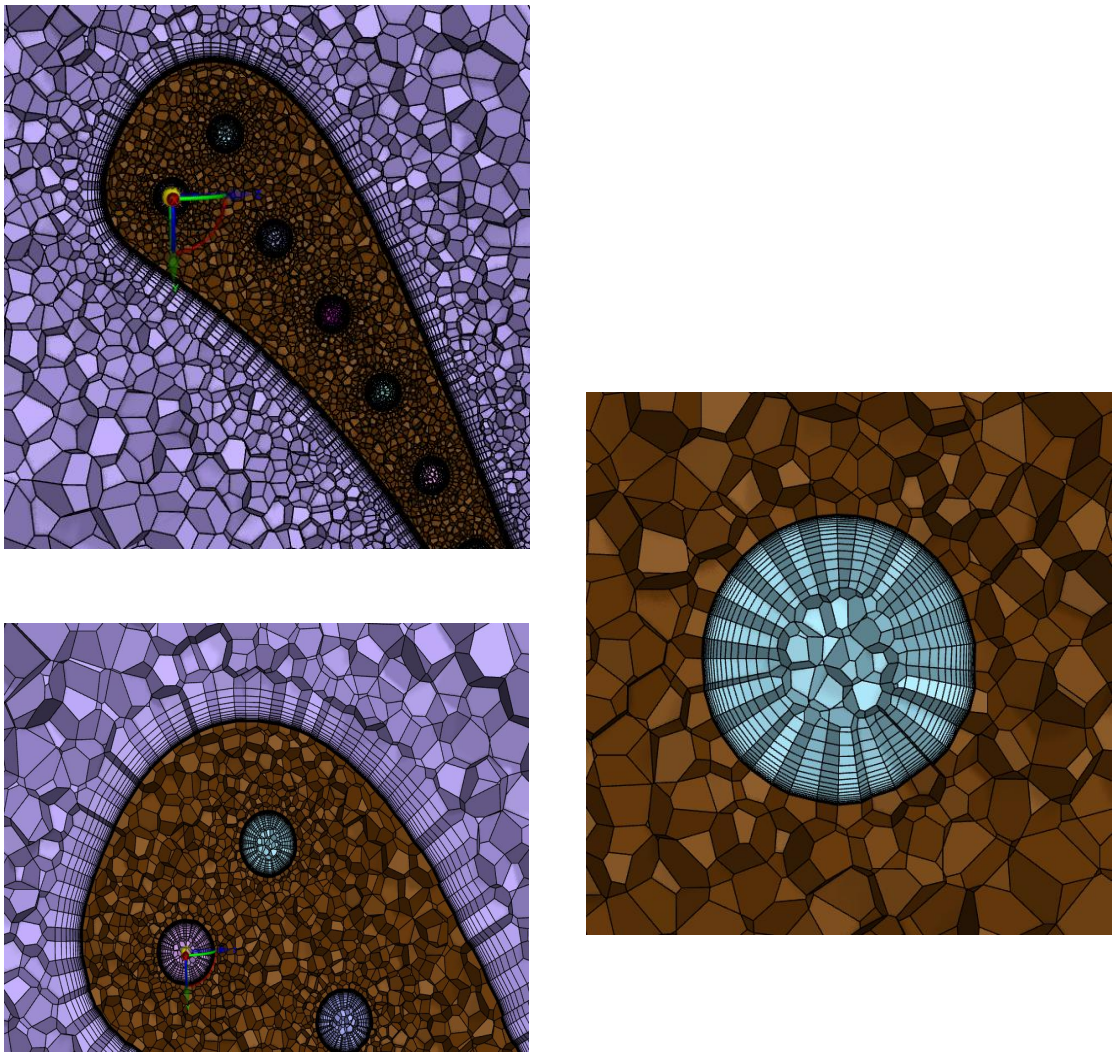
Reconstructed NASA C3X turbine vane geometry and periodic cascade computational domain.



The turbine vane geometry was independently created in ANSYS SpaceClaim using published experimental data to allow realistic CFD analysis of cascade aerothermal behaviour.

FIGURE 2

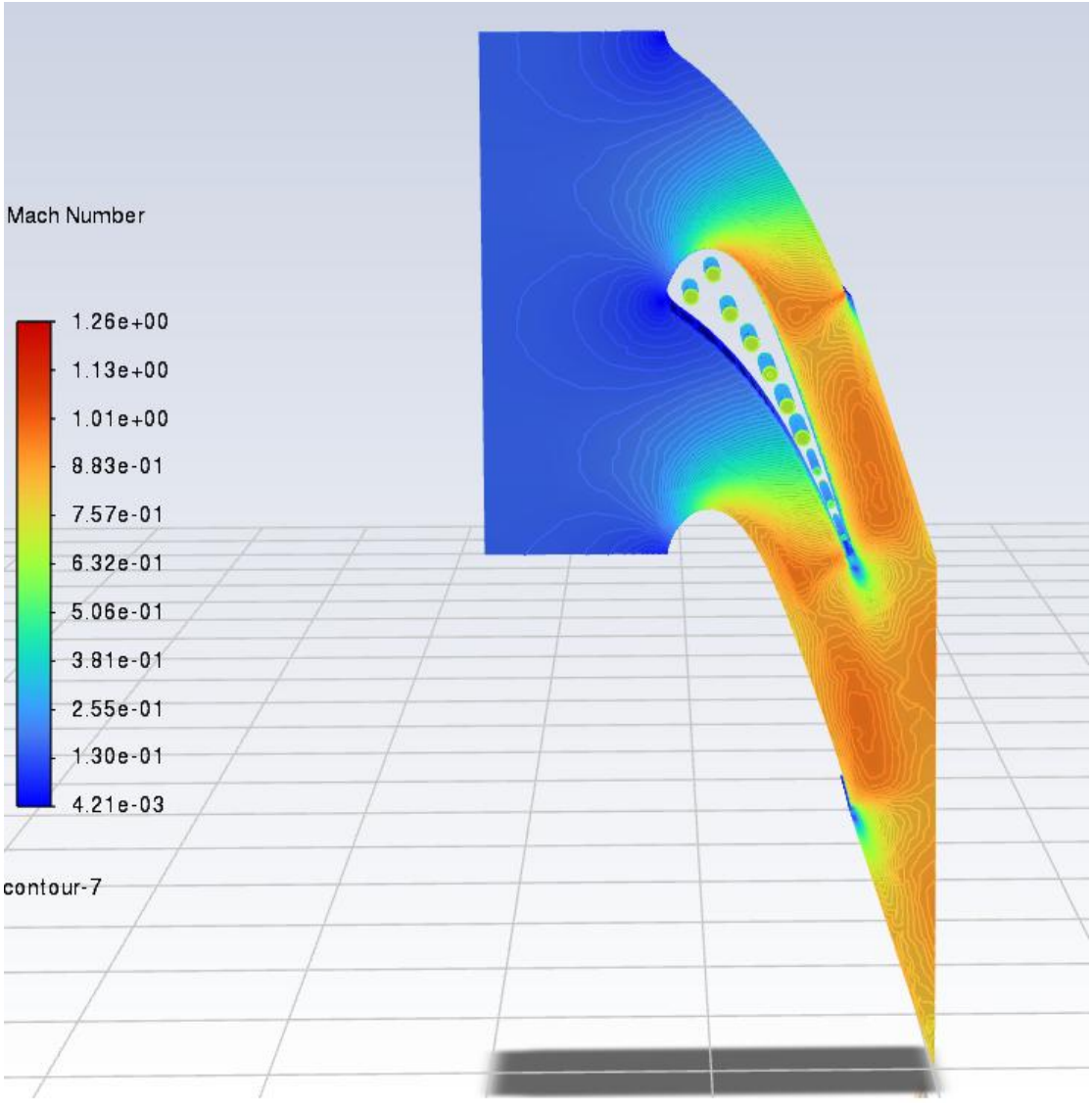
High quality computational mesh with boundary layer refinement near turbine vane surface and cooling hole region.



The mesh was carefully refined near the turbine vane surface and around the cooling hole regions to accurately capture near wall flow behaviour and heat transfer effects. Adequate boundary layer resolution was ensured to improve prediction of aerodynamic and thermal gradients. Mesh quality and sensitivity considerations were also evaluated to support reliable numerical convergence and validation of simulation results.

FIGURE 3

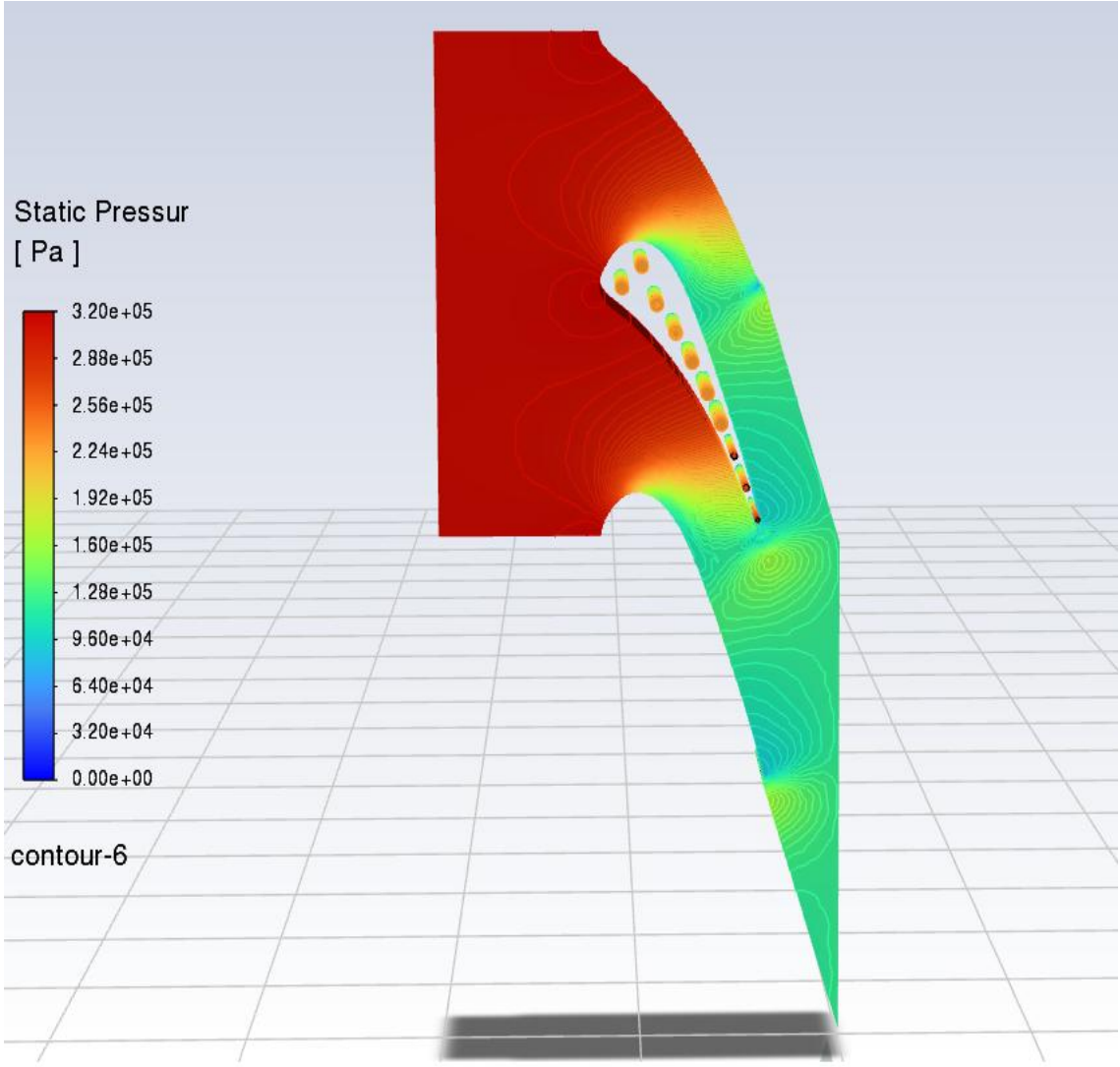
Mach number distribution showing flow acceleration along the turbine vane suction surface and interaction of cooling jets with the mainstream flow in the cascade passage.



The Mach number distribution shows how the flow accelerates along the turbine vane passage, especially near the suction surface where high speed regions are observed. The interaction between the main flow and the cooling jets creates local disturbances in the flow field, which influence the aerodynamic and thermal behaviour of the vane. These results help in understanding how cooling design affects turbine performance under high temperature operating conditions.

FIGURE 4

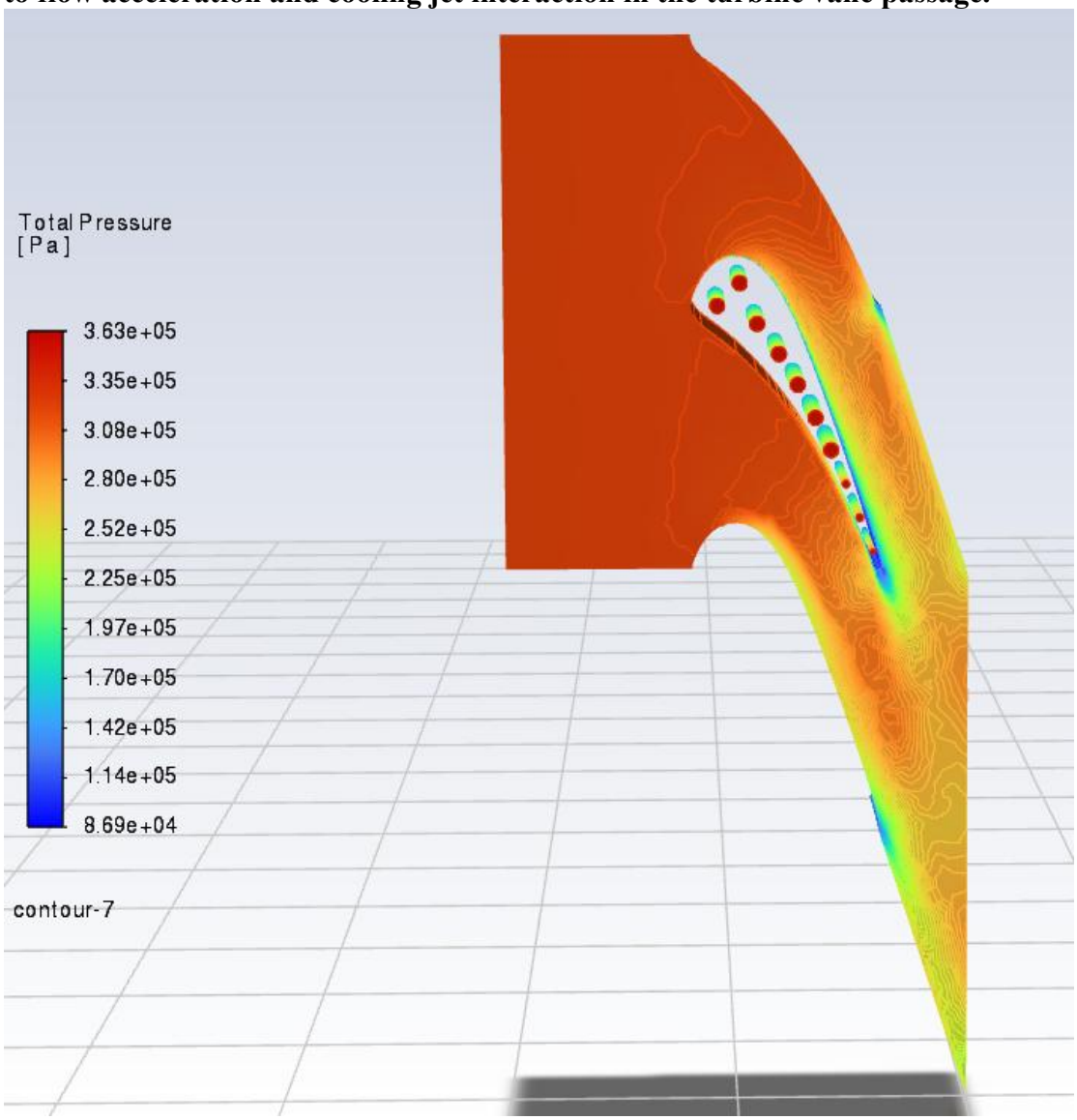
Static pressure distribution along the turbine vane passage showing pressure reduction on the suction surface and higher pressure region near the pressure side.



The static pressure contour shows how the pressure changes as the flow moves through the turbine vane passage. A clear pressure drop can be seen along the suction side due to flow acceleration, while higher pressure remains near the pressure side. This pressure difference is important for understanding the aerodynamic loading and overall flow behaviour in the turbine.

FIGURE 5

Total pressure distribution showing aerodynamic losses and wake formation due to flow acceleration and cooling jet interaction in the turbine vane passage.



The total pressure contour shows how energy losses develop as the flow moves through the vane passage. A noticeable drop in total pressure can be seen along the suction side and wake region due to flow mixing and cooling jet interaction. These losses are important when analysing turbine efficiency and aerodynamic performance.

This self-initiated validated CFD study reflects my strong motivation to pursue advanced research in turbomachinery aerothermal design and propulsion system performance analysis at postgraduate level.